SQL ASSIGNMENT 7

Subject: Database Management Systems (SQL) **Topic:** Aggregation & Grouping (GROUP BY & HAVING) **Status:** Completed

QUESTION 1

Create a table called **Orders** with columns: `order_id`, `user_id`, `payment_method`, and `amount`. Insert at least 8 sample records representing different users and payment methods.

```
CREATE TABLE Orders ( order_id INTEGER PRIMARY KEY, user_id INTEGER, payment_method TEXT, amount REAL ); INSERT INTO Orders (order_id, user_id, payment_method, amount) VALUES (101, 1, 'UPI', 450.00), (102, 2, 'Card', 250.00), (103, 1, 'Wallet', 150.00), (104, 3, 'UPI', 550.00), (105, 2, 'COD', 180.00), (106, 4, 'Card', 600.00), (107, 3, 'UPI', 320.00), (108, 1, 'Card', 400.00);
```

QUESTION 2

Count how many orders were placed using each `payment_method` in the `Orders` table.

```
SELECT payment_method, COUNT(order_id) AS total_orders FROM Orders GROUP BY payment_method;
```

QUESTION 3

Find the total amount spent by each `user_id` in the `Orders` table.

```
SELECT user_id, SUM(amount) AS total_spend FROM Orders GROUP BY user_id;
```

QUESTION 4

Show payment methods where the average order amount is greater than 300 using `GROUP BY` and `HAVING`.

```
SELECT payment_method, ROUND(AVG(amount), 2) AS avg_amount FROM Orders GROUP BY payment_method HAVING AVG(amount) > 300;
```

QUESTION 5

Explain the difference between `WHERE` and `HAVING` using example queries.

`WHERE` filters individual rows **before** grouping occurs, whereas `HAVING` filters aggregated groups **after** `GROUP BY` evaluation.

```
-- WHERE example: Filters raw orders with amount > 200 before grouping SELECT payment_method, COUNT(*) AS order_count FROM Orders WHERE amount > 200 GROUP BY payment_method; -- HAVING example: Groups all orders first, then filters payment methods with total revenue > 700 SELECT payment_method, SUM(amount) AS total_revenue FROM Orders GROUP BY payment_method HAVING SUM(amount) > 700;
```